MAKERWYS README version 5.133

by Pete Dowson © 2026

INSTRUCTIONS

First, please note that this version of MakeRunways is not suitable for use on Win7 based systems or earlier. It is also not intended for use with FS2004 (FS9) or earlier (but see below about that)***.***

It is aimed primarily at P3D, MSFS2020 and MSFS2024, though it will also work with FSX and FSX-SE on Windows 10. For MSFS2024, use the VFS Projector procedure in the dedicated section below.

For older system setups please use **MakeRunways version 4.70**, available in **Download Links – Useful Additional Programs**, on the FSUIPC support Forum.

For: Microsoft Flight Simulator ("MSFS2020")

Create a folder for MakeRunways, wherever you like on your PC. Place MakeRunways.exe into that folder, then right-click on it, select Properties – Compatibiliy – Run as administrator. (That way you won't need to specify "run as administrator" each time you run it).

You do not need the Lorby-Si program **LorbySceneryExport** for MSFS operation.

Then all you need to do is run the EXE (i.e. double click on it, or make a shortcut for your desktop). For MS-Store and Steam installs of MSFS, it will find your scenery files and extract the data for its files. Hopefully, it will also do this for DVD installations, though this has not been tested at time of writing.

Note that it will take over10 minutes to complete – there is a lot of data to scan! All the files with be generated in the MakeRunways folder you run the program from. One of the files it generates (for MSFS only) is **SceneryList.txt**, which simply lists the scenery folders being scanned in the order they will be scanned in (with a layer number), along with a note as to whether each is **Active** or **disabled**.

The pathnames for the sceneries there are partial – the owning folder is shown only once, in the first lines. Concatenating that with the individual path with give the complete path, but this may be too long for Windows to accept. In the generated Runways.xml file the path given is the full one, preceded by \\?\ which is supposed to make Windows accept it as a long path – but note that this doesn't seem to work for all Windows functions.

NOTE that many third-party airports installed via the MS Store are encrypted, and may be hidden so that MakeRwys cannot even see the BGL files which contain the data it normally extracts. Therefore the data provided by MakeRwys will, in these cases tend to be that of the default ("fs-base") scenery. This problem does not seem to apply to Community add-ons.

Thanks go to Matt Johnson on fsDeveloper.com for his invaluable help in sussing out the many changes in the BGL format and encoding since FSX days. This version of my program would still be months away from being released without that help.

For: Microsoft Flight Simulator 2024 (MSFS2024)

MakeRunways reads airport data from BGL files made accessible locally by the MSFS2024 VFS Projector. Keep MSFS2024 running and the projector active throughout the scan. The MSFS2020 instructions above describe a different setup.

Preparing and running a scan

1. Place the release executable in a dedicated, writable folder. Keep this separate from any MakeRunways installation used for another simulator, so that generated files are not overwritten.

2. Start MSFS2024 and enable Developer Mode. In the developer menu, select Tools > Virtual File System, then VFS Projector > Start. Wait for the projected scenery folders to become available before running MakeRunways.

3. Run MakeRunways from its dedicated folder. The scan must include the projected MSFS2024 scenery, rather than only the MSFS2020 package folders. Check the scanned folder list as described below. LorbySceneryExport is not required for this procedure.

4. Leave the simulator and VFS Projector running until MakeRunways reports completion. Scan time depends on the amount of accessible scenery and on file access speed.

Automatic source selection

In the normal MSFS scan, MakeRunways first looks for the Microsoft Store VFS projection at the following path. It adds the four-digit scenery subfolders it finds there; no additional command-line option is required for this scan.

%LOCALAPPDATA
%\Packages\Microsoft.Limitless_8wekyb3d8bbwe\LocalState\VFSProjection\scenery\

Checking the results

The generated files are written to the MakeRunways folder. Check SceneryList.txt for the scenery folders included in the scan and Runways.txt for the processing log. The output formats and runway inclusion filters described later in this guide also apply to MSFS2024.

Confirm that Runways.txt contains the message "MSFS2024 VFS Projection found and selected." If no suitable VFS folders are found, the program falls back to its legacy MSFS scan. Completion of that scan does not confirm an MSFS2024 scan.

If expected airport data is missing, first check that VFS Projector is active and that the relevant scenery appears in the scan. Retain Runways.txt and SceneryList.txt when reporting a problem; another run may overwrite them.

Availability and limitations

A scan can include only the scenery files accessible at the time it runs. Do not assume that every installed or streamed airport is available to the projector, or that all third-party airport data can be read. MakeRunways does not decrypt protected scenery files.

Regenerate the data after changing scenery when the consuming application needs an updated database. The generated files contain extracted airport data; they are not a copy of the scenery.

MSFS2024 helicopter data and identifiers

In addition to the existing output files, MSFS2024 scans provide dedicated CSV files for heliports and helicopter parking stands. These files complement the legacy helipads.csv output and keep the different types of helicopter-related scenery data separate.

heliports_msfs2024.csv

Lists MSFS2024 airport records without a conventional runway, including heliports identified by simulator codes longer than four characters. It records the full identifier, latitude, longitude, altitude in feet, magnetic variation, source BGL path, scenery name, airport name and source category.

The Source field is MSFS2024_AIRPORT_NO_RUNWAY. This identifies the selection used for the export; it does not imply that every exported record contains a separate, explicit helipad record.

helistands_msfs2024.csv

Lists helicopter stand candidates extracted from MSFS2024 TaxiwayParking records. Each entry includes the airport identifier, latitude, longitude, airport altitude in feet, true heading, radius in metres and diameter in feet. Additional fields retain the encoded parking number, type, name and suffix values for reference.

The Source field is MSFS2024_TaxiwayParking. These entries represent helicopter parking stands; they should not automatically be treated as the helipad objects listed in helipads.csv.

Identifiers longer than four characters

MSFS2024 airport and heliport identifiers are not restricted to the traditional four-character field used by older applications. The dedicated heliport CSV retains simulator identifiers of five to eight characters rather than truncating them to four. Such identifiers are simulator identifiers and are not necessarily official ICAO location indicators.

Applications importing these files must accept the complete identifier as text. Do not assume that every legacy output format or consuming application supports identifiers longer than four characters.

Output files and checks

Look for these CSV files beside the MakeRunways executable and check that their modification dates correspond to the scan being reviewed. The helistand file is written when matching parking records are encountered; a file left from an earlier scan does not establish that matching records were found in the latest scan.

runways.xml remains the existing airport/runway database described later in this guide. The dedicated helicopter CSV files provide additional data alongside it.

Acknowledgement

MSFS2024 support contributed by Giovanni Miduri (Gjanosh61), with AI assistance from OpenAI's ChatGPT for analysis, coding and documentation.

For: FSX, FSX-SE, Prepar3D v1 to Prepar3D v6 inclusive

Place the EXE into the relevant main FS folder. For FSX Steam Edition this will be in the Steam folder, under "steamapps\common".

For Prepar3D versions 3 -- 6 see the additional notes just below, before:

Double click on the MakeRwys.exe file to execute it. The data files will be extracted and placed in that folder. A full log of its actions will be created, called “runways.txt”.

MakeRunways will find the correct Scenery.CFG file automatically, based on the EXE name it finds there and its version number.

NOTE for Prepar3D versions 3, 4, 5 and 6

Those Prepar3D version 3, 4, 5 and 6 scenery installs abiding by the new L-M system for addons will NOT be seen by MakeRunways unless you copy the **LorbySceneryExport** program from the ZIP into the Prepar3D folder, to sit beside MakeRwys.exe.

MakeRunways will then automatically use the Lorby program to create a file called **MakeRwys_Scenery.cfg** in your P3D3/4/5/6 folder. This then lists it ALL of the scenery layers with correct layering. MakeRunways will use that file instead of the original.

NOTE that you should run MakeRunways “as administrator”. This is especially true on Win10. The best way is to mark it needing admin privileges in its (or its shortcut) “Properties – Compatibility” settings (right click on the EXE or the shortcut).
The special CFG file will be left in the main P3D folder for your own reference to the priority ordering of your scenery, something difficult to do from all the XML files.

For P3Dv4 amd above, if you want to use the other excellent facilities provided, get the full AddonOrganizer for P3D from **http://lorby-SI.weebly.com/downloads.html.** But only the included program is used by MakeRwys.

Thanks go to **Lorby-SI** for the use of these programs, and for making them Freeware!

For: FS2000, FS2002, FS2004

Please see the note at the start of this document: you may need to use MakeRunways version 4.70 for these versions of Flight Sim. The current version may be okay but you would need to be running on Windows 10.

Place the EXE into your main FS folder where the SCENERY.CFG file is located.

Double click on it to execute it. The data files will be extracted and placed in that folder. A full log of its actions will be created, called “runways.txt”.

LIST OF FILES PRODUCED

The files produced by MakeRwys.exe are:

Runways.txt

This is a text file showing the data analysed by the program in readable format. Use this to check on things when something odd seems to happen. you can find conflicting or overriding sceneries and so on this way. However, be warned. The file will typically grow to over 10 megabytes. You need to view it with a text editor capable of dealing easily with large files.

Runways.xml

An XML file providing a database of airports and their runways. (For MSFS lease see the note above regarding the filepaths provided in this file).

FStarRC.rws

This is a binary runways database used by my FStarRC program.

Airports.fsm

This is a binary airports database used by FS_Meteo, at present after re-naming to "Runways.rws".

Runways.csv

This is a comma-separated text database of all the runways, sorted, and it is in the particular format used by Radar Contact (all versionsup until the change to the following:

R4.csv

This is the same as the previous CSV file, but includes additional runway data added at the end of each line. This is used in Radar Contact 4 (before RC4.3) for more precise ATC operations.

R5.csv

This is the same as R4.csv but with runway closure indicators in an extra two fields - CL closed for landing and/or CT closed for take-off. This file is used by RCV4.3 and RC5. **R5.bin**

This is a binary version of the R5.csv file, with some extra fields. Details of the format are given below.

G5.csv

This contains information about Gates, Ramps, Parking places, along with airline lists where specified. It is used by later versions of Radar contact. **T5.csv**

This contains details of all taxi paths. It is used by later versions of Radar contact. **T5.bin**

This is a binary version of the T5.csv file. Details of the format are given below.

F5.csv and F5x.csv

These both list all COM frequencies and associated names, including the airport name. The F5 file contains 25kHz frequencies, converted from 8.33 ones where necessary. F5x contains the full xxx.xxx frequency including any 8.33 specific frequencies.

F4.csv and F4x.csv

These both list all COM frequencies and associated names, including the airport name. Note that if you want to use this with RC4 you must copy it to the Radar Contact 'data' subfolder. Make a safe copy of your original F4.CSV file first (I usually just rename it "f4orig.csv").

The F4 file contains 25kHz frequencies, converted from 8.33 ones where necessary. F4x contains the full xxx.xxx frequency including any 8.33 specific frequencies.

Note that MakeRwys matches frequencies as best it can, with the following extra "fiddles" to ensure a good range of frequencies:

1. Clearance Delivery is preferably met by FS type 14 "CD", but else is met by type 7 "CLEARANCE".
2. Multicom is met by FS type 2 "MULTICOM", but failing that by type 4 "CTAF".
3. Approach is from FS type 8 "APPROACH", but failing that from a second Type 9 "DEPARTURE" if there are more than one of the latter.
4. Departure is from FS type 9 "DEPARTURE", but failing that from a second Type 8 "ARRIVAL" if there are more than one of the latter.

Centre, FSS, AWOS and ASOS frequencies aren't used for this file.

Helipads.csv

This is a list of all the Helipads found, sorted into ascending ICAO order. Details of the fields are given below.

IMPORTANT NOTES ABOUT RUNWAY INCLUSION

RUNWAYS MUST HAVE “Start Positions”

Runways are only included in any of the files (except for being logged in the Runways.txt file)

if they are equipped with defined thresholds, or "start positions". Without those they may as well be omitted because the files then won't serve their purpose of assisting programs locateand land aircraft.

The whole data structure within MakeRunways, from which it produces these files, is predicated on this assumption.

MINIMUM RUNWAY LENGTH

By default MakeRunways imposes a minimum runway length of 1500 feet, otherwise runways are omitted from the data files. This is to eliminate so-called "ghost" runways being included -- very small runways provided only to allow AI Traffic to be directed better for landings and takeoffs.

If necessary you can override this value. Just use a command line parameter in the form:

/>n where n gives the number of feet to be considered the maximum for exclusion. Take care not to make this too small for fear of including those "ghosts", but if you really do want to see all, you can set />0.

WATER RUNWAYS

The runway lists will normally not contain any water runways. If you need these included just add this command line parameter:

/Water

If you want files containing ONLY water runways use

/WaterOnly

FLAGGING JETWAYS

Since version 4.85 there is an option to flag those gates with Jetways defined in Airport Facilities Data bgl files. The flag is simply “Jetway” in an extra field in the G5.csv file. See the format details below. Enable this option by:

/Jetway

on the command line.

UPDATING RADAR CONTACT's TRANSITION ALTITUDES FILE

If you are a user of Aivlasoft's EFB program for FSX/P3D, then part of the updated data you will have is a file with the correct Transition Altitudes for most airports in the world. MakeRunways can now automatically use that file to update Radar Contact's list too.

In order to do this, you must copy RC's "M4.CSV" file into the root FS folder, next to MakeRunways. Don't worry about EFB -- if that is correctly installed then MakeRunways will find it.

To tell MakeRunways to update M4.CSV add a command line parameter: /+T

(It is “T” for “Transition Altitude”).

When MakeRunways has finished simply copy the M4.CSV file back into RC's Data folder.

GATES FOR AIRLINE FILE

There is also an optional file, listing all the Gates for a specifically selected airline. To obtain this for your preferred airline, run MakeRwys.exe with a command line parameter such as (e.g.) /BAW

for the airline code BAW (British Airways).

If you are also using the />n and/or /+T parameters, those must come first.

The gates are listed in sorted order of Airport, but not of Gate name or number -- the Gates are in their scenery file order. The file produced will be an ordinary text file named <airline code> Gates.txt. e.g.

BAW Gates.txt

in the above case.

ASSUME “MakeRwys_Scenery.cfg” FILE ALREADY GENERATED

Programs other than AddonOrganizer may pre-generate this CFG file (SimStarterNG for example). To prevent AddonOrganizer being called upon to do it again, or to simply assume it isn’t needed, use this command line option:

/SSNG

VALUES PROVIDED IN EACH CSV FILE

Note: Values marked with * have extra formatting detail described in the Notes below.

RUNWAYS.CSV (ORIGINAL RUNWAYS FILE -- for very old versions of RC): ICAO,

Rwy*, Latitude*, Longitude*, Altitude*, HeadingMag, Length*, ILSfreq* **R4.CSV**

(RUNWAYS FILE for RCV4, until 4.3):

ICAO, Rwy*, Latitude*, Longitude*, Altitude*, HeadingMag, Length*, ILSfreq*, Width*,

MagVar, CentreLatitude, CentreLongitude, ThresholdOffset* **R5.CSV** (RUNWAYS FILE for RCV5):

ICAO, Rwy*, Latitude*, Longitude*, Altitude*, HeadingMag, Length*, ILSfreqFlags*, Width*, MagVar, CentreLatitude, CentreLongitude, ThresholdOffset*,Status*

R5.BIN see below

RUNWAYS.XML (Airports and runways for "It's Your Plane")

For each Airport:

> ICAO id, ICAOName, City, Longitude, Latitude, Altitude (feet), MagVar, Source BGL filepath, Scenery layer title.

And, within each Airport section, for each runway, all this, as available:

> Runway id, Len (feet), Hdg (Magnetic: add MagVar for True), Def (surface), ILSFreq (nnn.nn), ILSHdg (Mag), ILSid, ILSslope, Lat and Lon of threshold/start, Lat and Lon of FS's "Start" point (which may be to the side of the runway), ClosedLanding (TRUE or FALSE) and ClosedTakeoff (TRUE or FALSE), EndLights (NONE, LOW, MEDIUM or HIGH), CenterLights (NONE, LOW, MEDIUM or HIGH), CenterRed (TRUE or FALSE), Threshold offset, ILS name (first 31 chars only), VASI lights, VASI values (X/Z/Spacing/Pitch), Approach lights, Pattern direction and altitude.

G5.CSV (GATES for RCV5):

Without the /Jetway command line option:

> ICAO, GateName*, GateNumber, Latitude, Longitude, Radius*, HeadingTrue, GateType*, AirlineCodeList ...

With the /Jetway command line option there is an extra field between the GateType and the AirlineCodeList (if one is present). The field is either empty, or contains the word “Jetway”.

F5.CSV and F5x.CSV (COMMS FREQUENCIES)

ICAO, CommsType*, Frequency, "name of facility"

(if possible, also with an entry for Airport Name with CommsType=0 and Frequency=0)

F4. CSV and F4x.CSV (COMMS FREQUENCIES)

ICAO, Airport Name, ATISfreq, CDfreq, GroundFreq, TowerFreq, UnicomFreq, MulticomFreq, ApproachFreq, DepartureFreq

T5.CSV (TAXIWAYS for RCV5):

ICAO,TaxiwayName,MinimumWidthMetres,PointList ...

where PointList is a sequence of: Latitude,Longitude,TaxiType*,WidthMetres

T5.BIN see below

HELIPADS.CSV

ICAO, Latitude, Longitude, Altitude(ft), HeadingTrue, Length(ft), Width(ft), SurfaceType*, Flags*

DEFINITION OF VALUE FORMATS

Surface type is a number 0-23, see list below.

Concrete	0	Dirt	12
Grass	1	Coral	13
Water	2	Gravel	14
(Unknown)	3	Oil-treated	15
Asphalt 4	4	Mats	16
(Unknown)	5	Bituminous	17
(Unknown)	6	Brick	18
Clay	7	Macadam	19
Snow	8	Planks	20
Ice	9	Sand	21
(Unknown)	10	Shale	22
(Unknown)	11	Tarmac	23

Flags is a numerical value made up of one of these "types":

NONE	0
H	1
SQUARE	2
CIRCLE	3
MEDICAL	4

plus optionally 16 for "Transparent" and/or 32 for "Closed".
Rwy is nnn for runway number, then 0, 1=L, 2=R, 3=C, 4=W (water)

Runways designated N, NE, etc are denoted by runway numbers over 36, as follows:

37	= N-S
38	= E-W
39	= NW-SE
40	= SW-NE
41	= S-N
42	= W-E

43 = SE-NW
44 = NE-SW
45 = N
46 = W
47 = NW
48 = SW
49 = S
50 = E
51 = SE
52 = NE

Latitude is of the threshold, or AFCAD's "runway start"

Longitude is of the threshold, or AFCAD's "runway start"

Altitude is in feet

Length is in feet

ILSFreq is given as nnnnn for nnn.nn, or just 0 when no ILS

ILSFreqFlags are ILSFreq then optional B (backcourse), D (DME) G (Glideslope) equipped

Width is in feet

ThresholdOffset is in feet

Status is CT for Closed for Takeoff and/or ,CL for Closed for Landing

Radius is in metres

TaxiType is

0 = unknown
1 = Normal
2 = Hold short
3 = unknown
4 = ILS hold short
5 = Gate/Park (set by MakeRwys to denote arrival at a parking place)
6 = ILS hold short no draw

GateName is either omitted or one of

Dock	PkSE
Park	PkS
PkN	PkSW
PkNE	PkW
PkE	PkNW

GateType is:

0 = none
1 = Ramp GA
2 = Ramp GA Small
3 = Ramp GA Medium
4 = Ramp GA Large
5 = Ramp Cargo
6 = Ramp Military Cargo
7 = Ramp Military Combat
8 = Gate Small
9 = Gate Medium
10 = Gate Large
11 = Dock GA

CommsType is:

0 = Special entry with airport name, zero frequency
1 = ATIS
2 = MULTICOM
3 = UNICOM
4 = CTAF
5 = GROUND
6 = TOWER
7 = CLEARANCE
8 = APPROACH
9 = DEPARTURE
10 = CENTRE
11 = FSS
12 = AWOS
13 = ASOS
14 = CLEARANCE PRE-TAXI
15 = REMOTE CLEARANCE DELIVERY

BINARY FILE FORMATS

R5.BIN: binary format Runways file Record format:

```
struct {
        char chICAO[4];
        unsigned short wRwyNum;
        char chDesig; // L, C, R or space
        char chStatus[3]; // CT, CL, CTL or all zero
        unsigned short wSurface; // 0-23 as in New BGL format
        float fLatThresh;
        float fLongThresh;
        float fLatCentre;
        float fLongCentre;
        float fAltitude; // feet
        float fThrOffset; // feet
        float fHdgMag;
        float fMagVar;
        float fLength; // feet
        float fWidth; // feet
        float fILSfreq; // zero if none  char
        chILSflags[4];
        float fILShdg;
        char chILSid[8];
        float fILSslope;
} rbin;
```

T5.BIN: binary format Taxiways file Record format:

```
struct {
        DWORD dwNumPts;
        char chICAO[4];
        char chName[8]; // zero if not named
        float fMinWidth;

} tbin;

followed by dwNumPts x

struct {
        float fLat;
        float fLon;
        float fType;
        float fWidth; // to next point, =0 for last point

} tpt;
```

RUNNING MAKERUNWAYS SILENTLY

If an application wishes MakeRunways to run without any progress dialogue then it can start it with the command line parameter /+Q

Peter & John Dowson, October 2026